

DD_SUB-WF-RESUME-01-FLOW-WIK - WIK Resume BPMN

Developer implementation guide: DD_SUB-WF-RESUME-01-FLOW-WIK-DEV_IMPL_GUIDE-v1.2.md.

1. Purpose

This DD defines the WIK operator BPMN for subscription resume.

Process key:

```
sub-resume-wik
```

This BPMN implements the parent resume contract from DD_SUB-WF-RESUME-01-Subscription_Resume-v1.2.md for WIK.

The WIK resume flow handles all four resume triggers:

```
CUSTOMER_REQUESTED_RESUME
ADMIN_FORCE_RESUME
DUNNING_RESUME
SCHEDULED_AUTO_RESUME
```

The flow is:

```
validate -> PENDING_RESUME -> compute billing window -> emit billing intent -> optional reconnection fee payment -> reacti
```

The optional pay-first gate exists in the BPMN, but it is dormant for WIK v1.2 default config because `reconnection_fee_on_customer_resume = false`.

2. Ownership

This DD owns:

- WIK-specific resume BPMN process key.
- WIK happy-path BPMN sequence.
- WIK trigger-dependent BPMN branching.
- WIK dunning resume behavior.
- WIK scheduled auto-resume behavior.
- WIK optional reconnection fee gate.
- WIK worker-topic composition.
- WIK message catches.
- WIK use of `rules.subscription.resume.wik`.
- WIK resume examples.

This DD does not own:

- Generic resume API contract and trigger rules. See parent resume DD.
- Shared operation framework. See SUB-WF-FRAMEWORK-01.
- Worker contracts. See DD_SUB-WF-FRAMEWORK-01-WORKERS.
- Worker implementation code.
- SUB-LM, BIL, FUL, NOT, BIL-04, scheduler internals.
- Pause-history table definition. See pause/shared lifecycle persistence.

3. Required Runtime Configuration

3.1 Framework Operation Config

What this table is for:

subscription_operation_config tells the subscription workflow framework which BPMN process to start for an operator and operation kind.

When to use it:

Use it when the API receives a RESUME request and needs to resolve the WIK BPMN process key.

Required row:

```
INSERT INTO subscription_operation_config (
  operator_code,
  operation_kind,
  default_bpmn_process_key,
  operation_timeout_seconds,
  feature_flags,
  enabled,
  updated_by
)
VALUES (
  'WIK',
  'RESUME',
  'sub-resume-wik',
  120,
  '{}',
  TRUE,
  'migration'
);
```

Column meaning:

Column	Purpose	Sample	Meaning
operator_code	Operator scope.	WIK	This row applies to WIK.
operation_kind	Subscription operation.	RESUME	This row is used for resume requests.
default_bpmn_process_key	Camunda process key to start.	sub-resume-wik	Start the WIK resume BPMN.
operation_timeout_seconds	Framework operation timeout.	120	Resume should normally finish quickly unless waiting on payment or fulfillment.
feature_flags	Operator-specific flags.	{}	No WIK resume flow flags in v1.2.
enabled	Runtime enablement.	TRUE	API may start WIK resume.
updated_by	Audit actor.	migration	Seeded by migration.

3.2 WIK Resume Config

What this table is for:

subscription_resume_config stores operator-level resume behavior.

When to use it:

Load it before Drools validation and pass it as a fact into rules.subscription.resume.wik.

Required row:

```
INSERT INTO subscription_resume_config (
  operator_code,
  customer_self_service_enabled,
  reconnection_fee_on_customer_resume,
  dunning_resume_verify_debt_cleared,
  customer_notification_enabled,
  updated_by
)
VALUES (
  'WIK',
  TRUE,
  FALSE,
  TRUE,
  TRUE,
  'migration'
);
```

Column meaning:

Column	Purpose	Sample	Meaning
operator_code	Operator scope.	WIK	Policy applies to WIK.

Column	Purpose	Sample	Meaning
customer_self_service_enabled	Allows customer portal resume.	TRUE	WIK customers may self-resume eligible voluntary pauses.
reconnection_fee_on_customer_resume	Enables customer resume fee.	FALSE	WIK default does not charge reconnection fee in v1.2.
dunning_resume_verify_debt_cleared	Re-checks debt with billing.	TRUE	Dunning resume asks BIL-01 to confirm debt cleared.
customer_notification_enabled	Enables NOT-01 resume notification.	TRUE	Send customer notification after success.
updated_by	Audit actor.	migration	Seeded by migration.

4. BPMN Start Variables

The shared framework starts Camunda with these variables.

Variable	Purpose	Example
operationId	Links BPMN to subscription_operation.	subop_01HZ_KAMAU_RESUME
operatorCode	Operator scope.	WIK
subscriptionId	Subscription being resumed.	sub_01HZ_KAMAU_FIBER
operationKind	Always RESUME.	RESUME
customerId	Customer reference from SUB-LM.	cust_01HZ_KAMAU
resumeTrigger	Trigger requesting resume.	DUNNING_RESUME
resumeReasonCode	Business reason for close-out.	DEBT_CLEARED_VIA_MPESA
openPauseHistoryRowId	Open pause row to close.	pause_01HZ_KAMAU_SUSPNP
openPauseOriginIntent	Origin intent from open pause row.	SUSPEND_NP
billingIntentKind	Billing intent resolved by rules.	RESUME_NP_INTENT
requiresDebtClearedCheck	Whether BIL-01 must verify debt cleared.	true
requiresReconnectionFeePayment	Whether payment gate should run.	false
dunningCycleReference	BIL-04 reference for dunning resume.	bil04-resume-2026-06-22
adminForceResumeReason	Admin reason when force-resuming.	Emergency restoration approved
initiatingActorUserId	Actor user/service.	bil04-svc-account
initiatingActorRole	Actor role.	BILLING_INTERNAL
initiatingWorkflowKind	Upstream workflow.	BIL_04_DUNNING
initiatingWorkflowRef	Upstream reference.	bil04-resume-2026-06-22

5. Worker Topics Used

These are logical Camunda external-task topics, not separate deployments. One worker app may subscribe to many topics, but logs, retries, metrics, and contracts must remain visible per topic.

Worker topic	Why WIK resume uses it
drools-decide	Re-check status, trigger, role, origin-intent, fee, and debt rules.
sub-lm-put-pending-status	Move subscription to PENDING_RESUME.
sub-lm-compute-cycle-window	Compute billing-cycle context for BIL-01.
bil01-emit-intent	Ask BIL-01 to evaluate RESUME_INTENT or RESUME_NP_INTENT.
bil01-charge-pre-authorize	Optional reconnection fee payment gate.
bil01-confirm-charge	Optional confirmation after payment message.
ful-call-activate	Ask FUL-03 to reactivate service.
sub-lm-put-master-fields	Close pause-history row and write resume audit fields.
sub-lm-commit-state	Commit subscription to ACTIVE.
kafka-emit	Write resume success/failure event to outbox.
notification-send	Send resume notification without failing BPMN on notification failure.
subscription-operation-finalize	Set operation terminal state.

Worker topic	Why WIK resume uses it
bil01-revert-pending-charge	Compensation for pending reconnection fee charge.
sub-lm-revert-to-prior	Compensation to return subscription to SUSPENDED.

6. Happy Path BPMN

```

Start
-> update operation state VALIDATING
-> drools-decide
-> gateway: eligible and origin allowed?
    no -> resume rejected path
    yes
-> update operation state PENDING_STATE_FLIP
-> sub-lm-put-pending-status
-> sub-lm-compute-cycle-window
-> update operation state BILLING_CALL
-> bil01-emit-intent
-> gateway: requiresReconnectionFeePayment?
    yes -> bil01-charge-pre-authorize -> wait payment-confirmed -> bil01-confirm-charge
    no -> continue
-> gateway: DUNNING_RESUME debt cleared?
    no -> resume rejected path
    yes or not dunning -> continue
-> update operation state FULFILLMENT_CALL
-> ful-call-activate
-> update operation state AWAITING_FULFILLMENT_RESPONSE
-> wait fulfillment-completed
-> update operation state COMMITTING_FINAL_STATE
-> sub-lm-put-master-fields
-> sub-lm-commit-state
-> update operation state EMITTING_EVENT
-> kafka-emit SubscriptionResumed
-> notification-send
-> subscription-operation-finalize COMPLETED
-> End

```

7. Step Details

7.1 drools-decide

Inputs:

```

{
  "kjarPackage": "rules.subscription.resume.wik",
  "requestedDecisions": [
    "validate-preconditions",
    "validate-origin-intent-gating",
    "compute-reconnection-fee-applicability"
  ],
  "factsJson": "{...resume facts...}"
}

```

Expected output for WIK dunning resume:

```

{
  "decisionsJson": {
    "eligible": true,
    "originGatingAllowed": true,
    "billingIntentKind": "RESUME_NP_INTENT",
    "requiresDebtClearedCheck": true,
    "reconnectionFeeApplicable": false
  }
}

```

If eligible = false or originGatingAllowed = false, route to resume rejection.

7.2 sub-lm-put-pending-status

Inputs:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "toStatus": "PENDING_RESUME",
  "currentTransitionType": "RESUME",
  "currentTransitionReasonCode": "DUNNING_RESUME"
}

```

Output consumed later:

```
{
  "priorStatusSnapshot": {
    "statusCode": "SUSPENDED",
    "currentTransitionType": null,
    "currentTransitionReasonCode": null
  }
}
```

7.3 bil01-emit-intent

For dunning resume:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intentKind": "RESUME_NP_INTENT",
  "intentContext": {
    "resumeTrigger": "DUNNING_RESUME",
    "resumeReasonCode": "DEBT_CLEARED_VIA_MPESA",
    "openPauseHistoryRowId": "pause_01HZ_KAMAU_SUSPNP",
    "openPauseOriginIntent": "SUSPEND_NP",
    "pausedSince": "2026-06-15T02:00:18Z",
    "pauseDurationDays": 7,
    "dunningCycleReference": "bil04-resume-2026-06-22",
    "verifyDebtCleared": true
  },
  "cycleWindow": {
    "cycleStart": "2026-06-01",
    "cycleEnd": "2026-06-30"
  }
}
```

Expected WIK dunning result:

```
{
  "intentDecisionId": "bidec_01HZ_KAMAU_RESUME",
  "debtClearedConfirmation": true,
  "lineItems": [],
  "totals": {
    "net": 0,
    "currency": "KES"
  }
}
```

If debtClearedConfirmation = false, reject with DEBT_NOT_CLEARED and leave subscription suspended.

7.4 Optional Reconnection Fee Gate

The payment gate runs only if:

```
{
  "requiresReconnectionFeePayment": true
}
```

For WIK v1.2 default config this is normally false, so BPMN skips:

```
bil01-charge-pre-authorize -> payment-confirmed -> bil01-confirm-charge
```

Keep the gateway in the BPMN because future WIK policy may enable a reconnection fee for CUSTOMER_REQUESTED_RESUME.

7.5 ful-call-activate

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intent": "RESUME_SERVICE",
  "policy": "RESUME_FROM_SUSPENSION",
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_RESUME:resume"
}
```

FUL-03 reactivates the customer service, for example broadband authentication and voice profile. The BPMN waits for fulfillment-completed or fulfillment-failed correlated by fulfillmentCorrelationId.

7.6 sub-lm-put-master-fields

This step closes the open pause-history row and writes resume audit fields before the state is committed.

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "fieldsJson": {
    "last_status_changed_at": "2026-06-22T10:00:18Z"
  },
  "pauseHistoryCloseOut": {
    "pauseId": "pause_01HZ_KAMAU_SUSPNP",
    "actualResumeAt": "2026-06-22T10:00:18Z",
    "resumeReasonCode": "DEBT_CLEARED_VIA_MPESA",
    "resumeActorUserId": "bil04-svc-account",
    "resumeActorRole": "BILLING_INTERNAL",
    "reconnectionFeeBillableEventId": null,
    "adminForceResume": false,
    "adminForceResumeReason": null,
    "dunningCycleReferenceAtResume": "bil04-resume-2026-06-22"
  }
}
```

Important:

resumeReasonCode is the business reason from the request or system caller. It is not the same as resumeTrigger.

The worker implementation may call SUB-LM and pause-history persistence behind one topic. Do not write this row with ad hoc SQL inside the BPMN worker if the project has a dedicated repository/service for pause history.

7.7 sub-lm-commit-state

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "expectedFromStatus": "PENDING_RESUME",
  "toStatus": "ACTIVE"
}
```

Emit success only after this commit succeeds.

8. Message Catches

Message	Correlation key	Used when
payment-confirmed	billableEventId	Only if reconnection fee was pre-authorized.
fulfillment-completed	fulfillmentCorrelationId	FUL-03 successfully reactivated service.
fulfillment-failed	fulfillmentCorrelationId	FUL-03 failed or rejected reactivation.

Payment wait has a 7-day boundary timer. On timeout, BPMN compensates and finalizes the operation as FAILED.

9. Drools Package

Required package:

```
rules.subscription.resume.wik
```

Required decisions:

Decision	Purpose
validate-preconditions	Checks suspended status, open pause row, trigger, role, and scheduled auto-resume conditions.
validate-origin-intent-gating	Applies the trigger by origin_intent matrix from the parent DD.
compute-reconnection-fee-applicability	Decides whether the optional payment gate runs.

Example rule intent:

```
rule "WIK resume requires suspended subscription"
when
  $req: ResumeRequestFact()
  $sub: SubscriptionStateFact(subscriptionId == $req.subscriptionId, statusCode != "SUSPENDED")
then
  modify($req) {
    setEligible(false);
    setEligibilityReasonCode("INVALID_STATE_FOR_RESUME")
  }
}
```

```

end

rule "WIK dunning resume only resumes non payment suspension"
when
  $req: ResumeRequestFact(resumeTrigger == "DUNNING_RESUME", originGatingAllowed == null)
  $pause: PauseHistoryFact(subscriptionId == $req.subscriptionId, actualResumeAt == null, originIntent != "SUSPEND_NP")
then
  modify($req) {
    setOriginGatingAllowed(false);
    setOriginGatingReasonCode("NOT_A_DUNNING_SUSPENSION")
  }
end

rule "WIK scheduled auto resume requires scheduled time"
when
  $req: ResumeRequestFact(resumeTrigger == "SCHEDULED_AUTO_RESUME", eligible == null)
  $pause: PauseHistoryFact(subscriptionId == $req.subscriptionId, actualResumeAt == null, resumeScheduledAt == null)
then
  modify($req) {
    setEligible(false);
    setEligibilityReasonCode("NO_SCHEDULED_RESUME_TIME")
  }
end

rule "WIK reconnection fee only for configured customer resume"
when
  $req: ResumeRequestFact(resumeTrigger == "CUSTOMER_REQUESTED_RESUME")
  $cfg: OperatorConfigFact(reconnectionFeeOnCustomerResume == true)
then
  modify($req) {
    setReconnectionFeeApplicable(true)
  }
end

```

10. Compensation Paths

Failure point	BPMN compensation
Drools rejects request	Emit SubscriptionResumeRejected; finalize FAILED.
Pending status update fails	No committed mutation expected; emit rejected; finalize FAILED.
Billing intent fails	Run sub-lm-revert-to-prior; emit rejected; finalize FAILED.
Debt-cleared check fails	Run sub-lm-revert-to-prior; emit rejected with DEBT_NOT_CLEARED; finalize FAILED.
Payment timeout	Run bil01-revert-pending-charge; run sub-lm-revert-to-prior; emit rejected; finalize FAILED.
Fulfillment fails	Revert pending charge if any; run sub-lm-revert-to-prior; emit rejected; finalize FAILED.
Pause-history close-out fails	Route to manual recovery; network may already be active.
State commit fails	Route to manual recovery.
Kafka emit fails	Outbox retries. BPMN should not mark event delivered synchronously.
Notification fails	Log and continue. Notification is reliability-isolated.

11. Worked Example - Dunning Resume

Kamau was suspended for non-payment on 2026-06-15. His open pause-history row has:

```

{
  "pauseId": "pause_01HZ_KAMAU_SUSPNP",
  "originIntent": "SUSPEND_NP",
  "systemManaged": true,
  "dunningCycleReference": "bil04-dunning-wave-2026-06-15-L3",
  "outstandingDebtAmount": 3500.00,
  "outstandingDebtCurrency": "KES"
}

```

At 2026-06-22 09:55, Kamau pays. BIL-04 calls resume.

Request:

```

POST /api/subscriptions/sub_01HZ_KAMAU_FIBER/resume
Idempotency-Key: resume-sub_01HZ_KAMAU-bil04-payment-cleared-20260622
Content-Type: application/json

{

```

```

    "operatorCode": "WIK",
    "resumeTrigger": "DUNNING_RESUME",
    "resumeReasonCode": "DEBT_CLEARED_VIA_MPESA",
    "dunningCycleReference": "bil04-resume-2026-06-22",
    "notes": "Customer paid INV_01HZ KES 3,500 via M-Pesa",
    "initiatingWorkflowKind": "BIL_04_DUNNING",
    "initiatingWorkflowRef": "bil04-resume-2026-06-22"
  }

```

Runtime result:

```

API validates request
-> loads SUB-LM subscription summary: SUSPENDED
-> loads WIK resume config and open SUSPEND_NP pause row
-> Drools accepts DUNNING_RESUME on SUSPEND_NP
-> framework creates operation subop_01HZ_KAMAU_RESUME
-> Camunda starts sub-resume-wik
-> subscription moves SUSPENDED -> PENDING_RESUME
-> billing receives RESUME_NP_INTENT and confirms debt cleared
-> reconnection fee gate is skipped
-> FUL-03 reactivates service
-> pause-history row is closed with resumeReasonCode=DEBT_CLEARED_VIA_MPESA
-> subscription moves PENDING_RESUME -> ACTIVE
-> SubscriptionResumed event is emitted
-> notification is sent
-> operation is COMPLETED

```

After success:

- Subscription status is ACTIVE.
- Customer service is restored.
- Open pause-history row now has actual_resume_at.
- BIL-02 resumes cycle accrual from the resume event.
- BIL-04 closes the dunning loop.

12. Worked Example - Scheduled Auto Resume

Njeri paused her WIK subscription for 30 days. The pause row has:

```

{
  "pauseId": "pause_01HZ_NJERI_VAC",
  "originIntent": "CUSTOMER_REQUESTED_PAUSE",
  "resumeScheduledAt": "2026-07-01T00:00:00+03:00",
  "actualResumeAt": null
}

```

When resume_scheduled_at is due, the scheduler calls:

```

POST /api/subscriptions/sub_01HZ_NJERI_FIBER/resume
Idempotency-Key: scheduler-resume-pause_01HZ_NJERI_VAC-20260701
Content-Type: application/json

```

```

{
  "operatorCode": "WIK",
  "resumeTrigger": "SCHEDULED_AUTO_RESUME",
  "resumeReasonCode": "SCHEDULED_PAUSE_EXPIRED",
  "initiatingWorkflowKind": "FOUNDATION_SCHEDULER",
  "initiatingWorkflowRef": "pause_01HZ_NJERI_VAC"
}

```

Runtime result:

```

Drools confirms scheduler role and non-null resume_scheduled_at
-> origin-intent CUSTOMER_REQUESTED_PAUSE is allowed
-> billing receives RESUME_INTENT with no fee
-> payment gate skipped
-> FUL-03 reactivates service
-> pause-history row closes
-> subscription becomes ACTIVE

```

If scheduler accidentally targets SUSPEND_NP, reject with NOT_A_SCHEDULED_RESUME.

If scheduler targets a row with resume_scheduled_at = NULL, reject with NO_SCHEDULED_RESUME_TIME.

13. Cross-DD Dependencies

Dependency	Why it matters
DD_SUB-WF-RESUME-01-Subscription_Resume-v1.2.md	Owns generic resume API, trigger matrix, events, and invariants.
DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0.md	Owns operation creation and process start.
DD_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md	Owns worker-topic contracts.
DD_SUB-LM-01-Subscription_Lifecycle_Management-v1.0.md	Owns subscription status source of truth.
SUB-WF-PAUSE-01	Creates voluntary/admin pause rows that resume closes.
SUB-WF-SUSPEND-NP-01	Creates dunning suspension rows that dunning resume closes.
BIL-01	Owns resume billing intent, fee, and debt validation.
BIL-02	Consumes SubscriptionResumed to resume cycle accrual.
BIL-04	Calls DUNNING_RESUME and consumes result.
FUL-03	Owns network reactivation.
NOT-01	Owns customer notification.
FOUNDATION_SCHEDULER	Calls SCHEDULED_AUTO_RESUME for due pause rows.

14. Open Items

- WUG, WTZ, and YASSN resume flow satellites still require operator-specific deep dives.